

- 2 The simplified diagram in Fig 2.1 represents the trunk of a person bent forward, with the spine at an angle of 70.0° to the vertical. The extensor muscle, which joins the spine to the pelvis, makes an angle of 8.0° to the spinal column to maintain static equilibrium. The weight W of the person's trunk and head, which is 400 N, is shown to act at the point along the spine where the extensor muscle is attached.

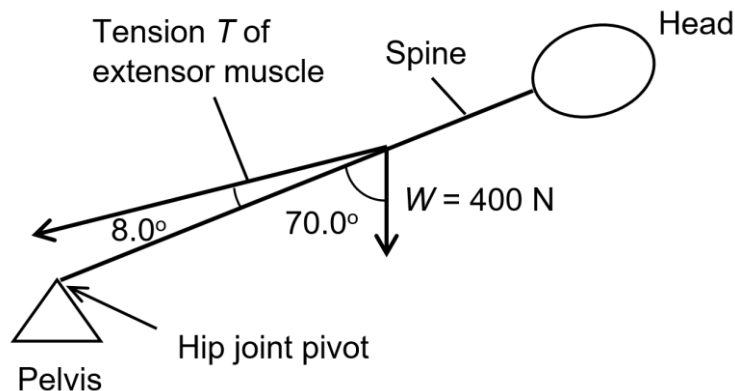


Fig 2.1

- (a) (i) Draw and label an arrow R on Fig 2.1 to show the direction of the reaction force acting on the spinal column at the hip joint pivot. [1]
- (ii) With reference to your answer in (a)(i), explain why there would be no net moments acting on the body.
- [1]
- (b) (i) Explain why the tension T of the extensor muscle, the weight W and the reaction force R must form a closed vector triangle.
- [1]

- (ii) Hence, or otherwise, calculate reaction force R at the hip joint and the tension T in the extensor muscle.

force $R = \dots\dots\dots$ N [2]

tension $T = \dots\dots\dots$ N [2]

- (c) Explain why it is advisable to lift heavy objects with the spine in a vertical position as compared to a bent position as shown in Fig 2.1.

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[Total: 9]